

Example 2: Fetch data from Statistics Estonia with R

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Setup R

```
library(quarto)      # for compiling Quarto presentations and documents
library(pxweb) # query and fetch data from Statistics Estonia
library(tidyverse)   # dplyr, ggplot2, readr, etc.
library(here)        # path settings
```

Accessing Statistics Estonia Data with pxweb

- ▶ Statistics Estonia is **not yet part of DBnomics**
- ▶ There is **no dedicated R package** for andmed.stat.ee
- ▶ **However**, you can access the data using the pxweb R package
- ▶ It is based on the **PX-WEB API**, also used by many Nordic statistical agencies

Data Access Options

▶ **Interactive Access**

Browse tables and build queries via the Statistics Estonia website

▶ **Direct Programmatic Query**

Use the API endpoint (a *URL*) and supply **key-value pairs** to specify:

- ▶ Variables (e.g., indicators, time, sector)
- ▶ Values or selections for each variable

Find and download data interactively

```
pxweb_interactive() # All available databases  
pxweb_interactive("https://andmed.stat.ee/api/v1/en/stat")
```

```
=====  
R PXWEB: Content of 'andmed.stat.ee'  
      at '/api/v1/en/stat'
```

```
=====  
[ 1 ] : Environment  
[ 2 ] : Economy  
[ 3 ] : Population  
[ 4 ] : Social life  
[ 5 ] : Multidomain statistics  
[ 6 ] : Population and Housing Census  
[ 7 ] : Discontinued datasets  
=====
```

Enter your choice:

```
('esc' = Quit, 'b' = Back, 'i' = Show id)
```

```
1:
```

GET call to PXWEB API

How to identify key-value pairs?

► Use either:

1. Programmatically via the metadata (pxweb_get())
2. From the Statistics Estonia web interface – click **“API query for this table”**

```
pa111_meta <- pxweb_get(  
  url = "https://andmed.stat.ee/api/v1/en/stat/PA111")  
pa111_meta
```

PXWEB METADATA

PA111: AVERAGE MONTHLY GROSS WAGES (SALARIES), MEDIAN, DEC
variables:

[[1]] Näitaja: Indicator

[[2]] Tegevusala: Economic activity

[[3]] Vaatlusperiood: Reference period

► Values for the first key (Näitaja)

```
pa111_meta$variables[[1]]
```

```
$code
```

```
[1] "Näitaja"
```

```
$text
```

```
[1] "Indicator"
```

```
$values
```

```
[1] "GR_W_AVG"      "NR_EMPL"      "GR_W_D1"      "GR_W_D2"  
[6] "GR_W_D4"      "GR_W_D5"      "GR_W_D6"      "GR_W_D7"  
[11] "GR_W_D9"      "GR_W_AVG_SM"
```

```
$valueTexts
```

```
[1] "Average monthly gross wages (salaries), euros"  
[2] "Number of employees"  
[3] "1st decile of monthly gross wages (salaries), euros"  
[4] "2nd decile of monthly gross wages (salaries), euros"  
[5] "3rd decile of monthly gross wages (salaries), euros"
```

Fetch data

- ▶ Set: Url to the database & Key-value pair
- ▶ Wildcard operator * to select all values

```
px_query_list1 <- list(  
  "Näitaja" = c("GR_W_D1", "GR_W_D5", "GR_W_D9"),  
  "Tegevusala" = "*",  
  "Vaatlusperiood" = "*"  
)  
  
pa111 <- pxweb_get(  
  url = "https://andmed.stat.ee/api/v1/en/stat/PA111",  
  query = px_query_list1  
)
```

- ▶ We get a list object with the data

```
pa111
```

PXWEB DATA

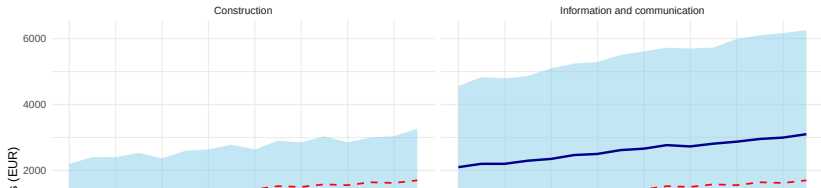
With 4 variables and 960 observations.

Convert the pxweb list object to a data frame

```
# Convert to data frame
pa111.df <- as.data.frame(pa111, column.name.type = "text")
pa111.code <- as.data.frame(pa111, column.name.type = "code")
# Optional: Combine text and code in one data frame
pa111.df$Näitaja <- pa111.code$Näitaja
```

```
ggplot(pa111_plotdata, aes(x = quarter)) +
  geom_ribbon(aes(ymin = D1, ymax = D9), fill = "skyblue",
  geom_line(aes(y = Median), color = "darkblue", linewidth
  geom_line(aes(y = total_median), color = "red", linetype
  facet_wrap(~`Economic activity`, ncol = 2) +
  labs(
    title = "Wage Distribution by Sector (1st to 9th Decile
    subtitle = "Red dashed line shows overall median (Total
    x = "Quarter",
    y = "Gross Nominal Monthly Wages (EUR)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Wage Distribution by Sector (1st to 9th Decile, Median)
Red dashed line shows overall median (Total – all activities)



Rendering All Formats from R or Bash

To render the document to **all output formats** (docx, Beamer, Reveal.js): run programmatically from another script or the R console:

```
quarto::quarto_render(here("teaching_materials", "example2_
```

or from the command line:

```
quarto render example2_statistics_estonia.qmd
```